

A Rule-Based Hydro–Solar–Battery Renewable Dispatch Simulation with SCADA-Inspired Analytics for a Jabalpur Case Study

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Abstract—This paper presents a compact Python-based educational simulation that integrates a synthetic Jabalpur-inspired solar photovoltaic (PV) resource, battery energy storage, and a hydropower reserve model derived from operational observations at the Rani Avanti Bai Hydel Power Station. The objective is to satisfy an hourly 24-hour demand curve while prioritising fossil-free resources and quantifying residual reliance on an illustrative fossil-grid fallback. A rule-based controller dispatches resources in the ordered sequence: solar-to-load, solar-to-battery charging, battery discharge, hydroelectric generation, and finally fossil fallback. The hydropower component retains SCADA-inspired reference values including 34.10 MW active-power capacity, 11 kV generator voltage, 50 Hz frequency, 39.85 m effective head, 97.50 m³/s water discharge, 56.25% gate opening and 166.60 rpm generator speed. For the bundled deterministic scenario, the model supplies 82.98% of the 1,061.00 MWh daily demand using renewable energy, dispatches 479.84 MWh from hydro, and requires 180.57 MWh of fallback energy during high-deficit hours, producing an illustrative operational-emissions proxy of 148,063.60 kg CO₂. The study demonstrates renewable integration, battery state-of-charge (SOC) modelling, dispatch analytics and reproducible technical reporting, while explicitly remaining an offline simulation rather than a real plant-control or compliance study.

Index Terms—hydropower, solar photovoltaic, battery energy storage system, renewable dispatch, state of charge, SCADA analytics, carbon footprint, Python simulation

I. INTRODUCTION

Renewable-energy portfolios increasingly require the coordinated use of variable generation, storage and dispatchable low-carbon resources. Solar PV can contribute substantial daytime energy but cannot serve demand during non-sunlight hours without complementary flexibility. Battery storage can shift a limited quantity of surplus solar production, while hydropower can provide controllable renewable reserve during deficits. This project develops a transparent minimum viable product (MVP) for studying those interactions in an offline environment.

The work combines two preceding portfolio prototypes created by the author: a solar–battery energy model for Jabalpur and a SCADA-inspired operational monitoring model based on observations made during hydropower vocational training [1], [2]. The integrated code is intended for GitHub publication as a reproducible software artefact that illustrates system modelling

and data interpretation, rather than as a representation of a utility dispatch centre.

The contributions of this paper are: (i) a simple modular dispatch algorithm that preferentially uses solar and stored solar before calling hydropower; (ii) integration of SCADA-inspired hydro reference parameters into renewable-dispatch analytics; (iii) explicit carbon-footprint accounting for non-renewable fallback; and (iv) a Python implementation that generates CSV, JSON and graphical outputs from a single executable cell or script.

II. SAFETY AND SCOPE BOUNDARY

The simulation is strictly read-only and offline. It does not connect to a SCADA network, programmable logic controller, protection relay, turbine governor, spillway or intake gate, PV inverter, battery-management system, breaker, meter, or live electrical grid. The hydropower measurements are treated as contextual observations and model reference values only. Alert labels and estimated gate-opening indications are educational analytics, not authorised plant alarm settings or operating instructions.

All capacities and the daily demand profile form a fictional regional scenario inspired by Jabalpur and the author’s training exposure. Consequently, the computed outputs quantify the behaviour of the simulation assumptions, not the operational performance, approval status, or environmental impact of the actual Rani Avanti Bai facility.

III. SYSTEM ARCHITECTURE AND INPUT DATA

A. Integrated Energy Architecture

The model comprises four supply elements: a synthetic solar PV generator, a battery energy storage system (BESS), an available hydropower reserve and an illustrative fossil-grid fallback. An hourly demand profile is supplied to a rule-based energy manager. Solar energy serves demand first; excess solar may charge the battery; stored solar is dispatched against later deficit; hydro is used only after solar and available battery output are insufficient; and fallback energy completes the balance if the renewable resources reach their limits.

TABLE I
INTEGRATED SIMULATION CONFIGURATION

Parameter	Value
Simulation horizon / interval	24 h / 1 h
Synthetic PV maximum capacity	55.00 MW
Battery capacity	70.00 MWh
Battery initial SOC	45.00%
Battery minimum–maximum SOC	10.00–90.00%
Battery charge/discharge limit	16.00 MW
Charge/discharge efficiency	95.00% / 95.00%
Hydro maximum reference output	34.10 MW
Fossil fallback emission factor	820 kg CO ₂ /MWh

TABLE II
HYDROPOWER REFERENCE OPERATING VALUES

SCADA-inspired signal	Reference value
Active power output	34.10 MW
Generator voltage	11.00 kV
Frequency	50.00 Hz
Reservoir level	409.55 m
Effective head	39.85 m
Water discharge	97.50 m ³ /s
Gate opening	56.25%
Generator speed	166.60 rpm

B. Primary Configuration

Table I records the parameters embedded in the final Python code. Power is expressed in MW and each one-hour interval is therefore numerically equivalent to MWh for energy accounting.

C. SCADA-Inspired Hydropower Reference

The hydro reserve capacity and monitoring context are derived from the author's SCADA imitation input, based on an observed control-room reading. Table II preserves these values in the integrated program [2]. The code uses active power as the hydro output ceiling; the remaining parameters are displayed as reference monitoring values. An estimated gate-opening field is calculated proportionally from simulated hydro loading only for visual analytics.

IV. MATHEMATICAL MODEL AND CONTROL LOGIC

A. Solar Availability

A deterministic solar profile avoids dependence on internet access or third-party datasets in the MVP. At hour t , available PV output is:

$$P_{PV,t} = P_{PV,\max} s_t c_t, \quad (1)$$

where $P_{PV,\max} = 55$ MW, s_t is a clipped sinusoidal daylight shape, and c_t is a fixed cloud modifier. This representation produces zero output at night and a midday maximum.

B. Battery State of Charge

The BESS stores only unused solar generation. For one-hour intervals, charging and discharging are governed by:

$$E_{t+1} = E_t + \eta_c P_{c,t} \Delta t - \frac{P_{d,t} \Delta t}{\eta_d}, \quad (2)$$

subject to:

$$E_{\min} \leq E_t \leq E_{\max}, \quad (3)$$

$$0 \leq P_{c,t} \leq 16 \text{ MW}, \quad (4)$$

$$0 \leq P_{d,t} \leq 16 \text{ MW}, \quad (5)$$

where $\eta_c = \eta_d = 0.95$, $E_{\min} = 7$ MWh and $E_{\max} = 63$ MWh.

C. Dispatch Rule

For each hourly demand D_t , the controller follows the priority sequence:

$$\text{PV} \rightarrow \text{BESS} \rightarrow \text{Hydro} \rightarrow \text{Fallback}. \quad (6)$$

Solar first supplies demand; surplus is offered to BESS charging. For any residual deficit, battery discharge is attempted, followed by hydro dispatch up to the available hydro curve, and finally non-renewable fallback. The hourly balance is:

$$D_t = P_{PV,\text{load},t} + P_{d,t} + P_{H,t} + P_{F,t}. \quad (7)$$

The order deliberately reflects the project request that hydro be used as renewable reserve after solar and stored solar have been utilised. It is not asserted to be an economic or reservoir-optimal operating policy.

D. Carbon Footprint Proxy

Solar, stored solar and hydro are assigned zero direct operational emissions for this simplified exercise. Carbon emissions arise only from fallback energy:

$$C_t = P_{F,t} \Delta t \gamma_F, \quad (8)$$

where $\gamma_F = 820$ kg CO₂/MWh. This is an illustrative modelling parameter, not a measured grid-emissions factor.

V. PYTHON IMPLEMENTATION

The code is written as a single Python page designed for Google Colab, Jupyter Notebook or local Python 3 execution. The only required third-party dependencies are NumPy, pandas and Matplotlib. The implementation separates the principal concerns into: (i) a configuration data class; (ii) deterministic solar and hydro availability functions; (iii) a Battery class for SOC transitions; (iv) an hourly dispatch loop; (v) metric calculation; and (vi) output generation.

The program generates three machine-readable result files: `hourly_hybrid_dispatch.csv`, `metrics_summary.json`, and `assumptions.json`. It also writes four PNG figures for repository documentation. The final combined source file is identified as `solar_hydro_battery_SINGLE_CELL_READY_TO_COPY.py` [3].

TABLE III
RULE-BASED DISPATCH PSEUDOCODE

For each hour t : obtain D_t , PV availability and hydro availability; serve demand from PV; charge BESS from PV excess within SOC and charge-power bounds; discharge BESS against remaining deficit; dispatch hydro against any remaining deficit; dispatch fossil fallback only if necessary; compute emissions and store SCADA-inspired analytics.

TABLE IV
KEY PERFORMANCE INDICATORS FROM THE FINAL CODE RUN

Indicator	Result
Daily electrical demand	1,061.00 MWh
Solar directly supplied to load	347.31 MWh
Solar offered to BESS charging	33.24 MWh
Battery delivered to load	53.28 MWh
Hydro energy dispatched	479.84 MWh
Fossil-proxy fallback energy	180.57 MWh
Renewable supply share	82.98%
Operational emissions proxy	148,063.60 kg CO ₂
Final battery SOC	7.00 MWh (10.00%)
Hours with hydro dispatch	18 h
Hours with fallback dispatch	8 h

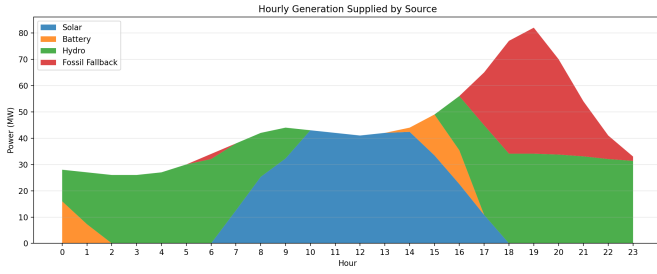


Fig. 1. Hourly demand supply by solar, battery, hydro and fossil-proxy fallback.

VI. SIMULATION RESULTS

A. Daily Energy Balance

The deterministic 24-hour run supplies a total demand of 1,061.00 MWh. Table IV summarises the resulting source allocations. Solar supplies 347.31 MWh directly to demand and charges 33.24 MWh into storage. Due to round-trip efficiency and SOC limits, the battery delivers 53.28 MWh across the day, which includes energy initially stored in the BESS at the beginning of the horizon. Hydro provides 479.84 MWh as reserve energy. The remaining 180.57 MWh is met by the illustrative fallback source.

B. Hourly Source Dispatch

Fig. 1 shows the source mix delivered to load. Daylight PV reduces hydro utilisation during the middle of the day. At hours 10–13, solar supplies the entire demand and supports charging of the BESS. In the evening, declining PV output coincides with a high demand peak, requiring both hydro and fallback energy.

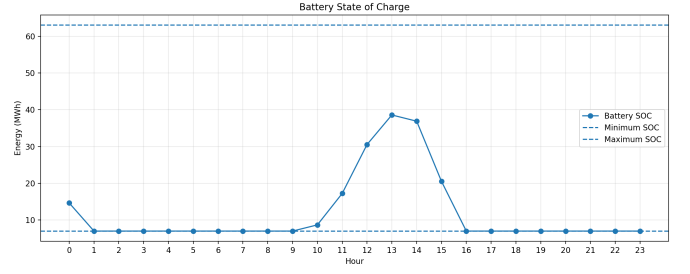


Fig. 2. Battery SOC trajectory subject to minimum and maximum operating bounds.

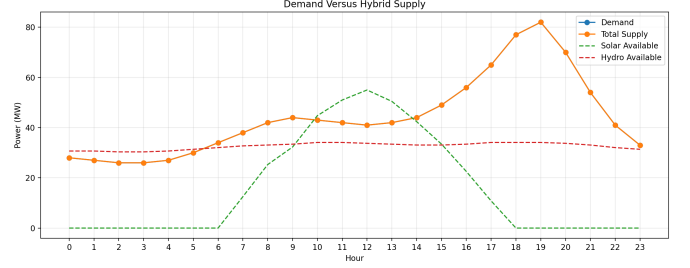


Fig. 3. Hourly demand, supplied power and available renewable resources.

C. Battery SOC Behaviour

Fig. 2 demonstrates the storage-shifting function. The battery begins at 45% SOC and discharges during early-hour demand deficits until it reaches the 10% reserve boundary. Daytime PV surplus then restores SOC before evening demand again exhausts the usable stored energy. This trajectory makes two findings clear: storage mitigates deficits, but the selected 70 MWh capacity and 16 MW discharge rating are insufficient to eliminate evening fallback for the selected demand profile.

D. Demand Adequacy and Renewable Availability

Fig. 3 overlays demand and total dispatched supply, along with renewable availability. Total supply matches demand in every hour because the final fallback resource closes unresolved deficits. The figure also indicates that the high evening demand exceeds the available hydropower reserve after PV output has fallen to zero and battery SOC has reached its lower boundary.

E. Carbon-Footprint Illustration

Because operational emissions are assigned only to the fallback source, Fig. 4 concentrates emissions in the deficit periods. Fallback appears at hour 6 and primarily across hours 17–23, with the highest carbon contribution at hour 19 when demand reaches 82 MW while PV availability is zero and hydro is constrained by the 34.10 MW reference capacity.

VII. ENGINEERING INTERPRETATION

The results show a complementary operating pattern. Solar energy provides valuable daytime supply and enables limited storage recovery. The BESS provides temporal shifting but reaches its minimum SOC at critical intervals. Hydropower materially reduces non-renewable fallback after solar and

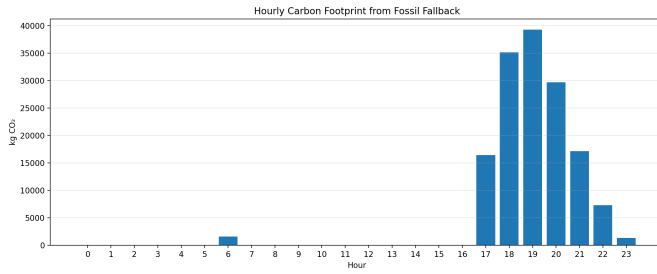


Fig. 4. Hourly illustrative CO₂ emissions due to fossil-proxy fallback dispatch.

storage limits are reached, demonstrating the flexibility value of dispatchable renewable resources. However, the non-zero fallback result also reveals a capacity adequacy issue under the assumed evening demand peak.

The final metrics should not be interpreted as an investment recommendation or grid-planning conclusion. The demand curve is synthetic; the PV output is an embedded illustrative curve rather than measured meteorological output; hydro availability is represented with a simplified factor profile; and the carbon factor is a teaching input. A rigorous engineering extension would replace those assumptions with time-series data, reservoir and turbine constraints, dynamic tariffs, lifecycle battery ageing, and power-flow validation.

VIII. GITHUB REPRODUCIBILITY PACKAGE

For repository presentation, the report should be stored under a `reports/` directory, the Python source under `src/` or the repository root, and automatically generated evidence under `outputs/figures/` and `outputs/data/`. A minimal repository structure is:

```
project-root/
  README.md
  solar_hydro_battery_SINGLE_CELL_READY_TO_COPY.py
  reports/
    Oihika_Arpit_IEEE_Hydro_Solar_Battery_MVP_Report.pdf
  outputs/
    hourly_hybrid_dispatch.csv
    metrics_summary.json
    assumptions.json
    figures/*.png
```

The code is reproducible without external APIs: a reviewer installs `numpy`, `pandas` and `matplotlib`, executes one Python file, and regenerates all reported figures and output data. This design is appropriate for an academic portfolio because it allows a university reviewer to trace model assumptions, dispatch decisions and KPI calculations.

IX. LIMITATIONS AND FUTURE WORK

The principal limitations are the use of synthetic input profiles, a one-day time horizon, a rule-based rather than optimized dispatch controller, and the absence of hydrological scheduling constraints. The SCADA parameters are contextual reference inputs rather than continuous measured signals. No

voltage control, protection, reactive-power behaviour, grid-code conformance, or equipment certification is modeled.

Future work should use archived solar weather data for Jabalpur, incorporate verified hydropower operational datasets with authorised units, extend the simulation to seasonal horizons, and compare rule-based dispatch with linear or mixed-integer optimisation. A future dashboard can integrate battery SOC, hydro operating parameters, load and carbon indicators in a visual monitoring interface while preserving the read-only simulation boundary.

X. CONCLUSION

This project provides an auditable Python MVP for renewable-resource coordination inspired by field training and solar-battery modelling. By combining synthetic PV generation, constrained battery storage and SCADA-inspired hydropower reserve dispatch, the simulation communicates how complementary renewable technologies can reduce dependence on carbon-emitting fallback supply. The resulting figures and machine-readable evidence make the work suitable for GitHub publication and renewable-energy academic portfolio use, provided that its educational and non-operational boundary remains explicit.

REFERENCES

- [1] O. Arpit, "Solar PV-Battery Hybrid Energy Simulation: Jabalpur, Madhya Pradesh, India," Python prototype and simulation assumptions, May 2026, unpublished portfolio artefact.
- [2] O. Arpit, "Rani Avanti Bai Hydel Power Station Operation Monitoring Dashboard," SCADA-inspired Python prototype with observed control-room reference values, May 2026, unpublished portfolio artefact.
- [3] O. Arpit, "Jabalpur Solar + Rani Avanti Bai Hydel + Battery MVP: Single-Cell Copy-Paste Python Program," source-code artefact accompanying this report, May 2026.